DIFFERENTIAL CALCULUS

Chapter I: Problems on Limits, Continuity and Differentiability

1. A function $f(x)$ is defined as follows:

$$f(x)=\begin{cases} 6+4x, & -\frac{3}{2}\le x<0 \\ 6-4x, & 0\le x<\frac{3}{2} \\ 6+4x, & x\ge\frac{3}{2}. \end{cases}$$

Discuss the continuity at $x=\frac{3}{2}$ and differentiability at $x=0.$ Also sketch the graph of $f(x).$

2. A function $f(x)$ is defined by

$$f(x)=\begin{cases} 2x-1, & x\le 1 \\ x^2-x+1, & x>1. \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x=1.$ Also sketch the graph of $f(x).$

3. A function $f(x)$ is defined by

$$f(x)=\begin{cases} 1+x, & x\le 0 \\ x, & 0<x<1 \\ 2-x, & 1\le x\le 2 \\ 2x-x^2, & x>2. \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x=1$ and $x=2.$ Also sketch the graph of $f(x).$

4. A function $f(x)$ is defined by

$$f(x)=\begin{cases} \frac{|x-3|}{x-3}, & x\ne 3 \\ 0, & x=3. \end{cases}$$

Find $\lim_{x\to 3} f(x),$ if it exists.

5. A function $f(x)$ is defined by

$$f(x)=\begin{cases} x, & 0\le x<1 \\ 2-x, & 1\le x\le 2 \\ x-\frac{1}{2}x^2, & x>2. \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x=2.$ Also sketch the graph of $f(x).$

6. A function $f(x)$ is defined by

$$f(x)=\begin{cases} x^2+1, & 0<x\le 1 \\ 4x^2-3x, & 1<x<2 \\ 3x+4, & x\ge 2. \end{cases}$$

Discuss the continuity of $f(x)$ at $x=1$ and the existence $f'(x)$ at $x=2.$ Also sketch the graph of $f(x).$

7. A function $f(x)$ is defined by

$$f(x)=\begin{cases} x^2+1, & x\le 0 \\ 2x-1, & 0<x\le 1 \\ x^2-x+1, & x>1. \end{cases}$$

Test the continuity and differentiability of $f(x)$ at $x=0$ and at $x=1.$ Also sketch the graph of $f(x).$

8. A function $f(x)$ is defined by

$$f(x)=\begin{cases} x^2\sin\dfrac{1}{x}, & x>0 \\ -x^2, & x\le 0. \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x=0.$

9. Consider the function

$$f(x)=\begin{cases} 2x+5, & -3\le x<0 \\ -3, & x=0 \\ -5x^2, & x>0. \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x=0.$

10. Consider the function

$$f(x)=\begin{cases} x^2, & x\le 0 \\ x, & 0<x<1 \\ \dfrac{1}{x}, & x\ge 1. \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x=0$ and $x=1.$

Differentiation Explicit and Implicit Functions

1. If $f(x)=\ln\dfrac{\sqrt{a+bx}-\sqrt{a-bx}}{\sqrt{a+bx}+\sqrt{a-bx}},$ find for what values of $x,$ $\dfrac{1}{f'(x)}=0.$

2. Find $\dfrac{dy}{dx},$ where $y=\dfrac{(x^3-2x+5)^4\sqrt[3]{x^2+3x+2}}{x^2\tan\frac{1}{2}x}.$

3. If $f(x)=\left(\dfrac{a+x}{b+x}\right)^{a+b+2x},$ show that $f'(0)=\left(2\log\dfrac{a}{b}+\dfrac{b^2-a^2}{ab}\right)\left(\dfrac{a}{b}\right)^{a+b}.$

Chapter II: Indeterminate Forms, L' Hospital's rule

Evaluate the following:

(i) $\lim_{x\to 0} \frac{x+\sin 2x}{x-\sin 2x}$ (ii) $\lim_{x\to 0} \left(\frac{4}{x^2} - \frac{2}{1-\cos x}\right)$ (iii) $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$

(iv) $\lim_{x\to 0} \frac{x-\sin^{-1} x}{\sin^3 x}$ (v) $\lim_{x\to \frac{\pi}{2}} (\sin x)^{\tan x}$ (vi) $\lim_{x\to 0} \frac{x-\sin x}{\tan^3 x}$

(vii) $\lim_{x\to 0} (a^x + x)^{\frac{1}{x}}$ [**Hints:** $\frac{d}{dx}(a^x) = a^x \log_e a,$ where a is a positive constant]

(viii) $\lim_{x\to 0} \frac{\ln \ln(1-x^2)}{\ln \ln \cos x}$ (ix) $\lim_{x\to \frac{\pi}{4}} \frac{\sec^2 x - 2\tan x}{1+\cos 4x}$ (x) $\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$

(xi) $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}}$ (xii) $\lim_{x\to 0} \frac{(e^x-1)\tan^2 x}{x^3}$ (xiii) $\lim_{x\to 0} (1+x)^{\frac{1}{x}}$

(xiv) $\lim_{x\to 0} \left(\frac{1}{x} - \frac{1}{\sin x}\right)$ (xv) $\lim_{x\to 0} \left(\frac{1}{\ln x} - \frac{x}{x-1}\right)$ (xvi) $\lim_{x\to 0} \frac{xe^x - \sin x}{3x^2}$

(xvii) $\lim_{x\to 0} \frac{x-\sin x}{x^3}$ (xviii) $\lim_{x\to 0} \frac{\sin x - \ln(e^x \cos x)}{x\sin x}$ (xix) $\lim_{x\to 0} (x)^{\frac{1}{1-x}}$.

Chapter III: Expansion of Functions

1. State Taylor's theorem and Maclaurin's theorem. Use Maclaurin's theorem to expand $e^{2x\sin 3x}$ in a power series.
2. Expand $\tan\left(x+\frac{\pi}{4}\right)$ in a series of ascending powers of x up to fourth powers of x.
3. State Rolle's theorem. Give the geometrical interpretation of Rolle's theorem and Mean value theorem.
4. Verify Rolle's theorem for $f(x) = e^{-x}\sin x$ in the interval $(0,\pi)$.
5. Give the statement of Taylor's theorem. Expand $\cos^2 x$ in powers of $x-\frac{\pi}{2}$.
6. Give the statement of Maclaurin's theorem.
   Expand $\frac{x}{e^x-1}$ in a series of ascending powers of x.
7. State Lagrange's Mean value theorem. Find the value of x_0 in the Mean value theorem for the function $f(x) = x^3 - 5x^2 + 12x - 20$ in the interval (0,4).
8. Use the Mean Value theorem to show
$$\frac{x}{1+x} < \ln(1+x) < x \text{ for } -1 < x < 0 \text{ and for } x > 0.$$
9. Given $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x+\theta h), 0<\theta<1,$ where $f(x) = (x-a)^{\frac{5}{2}}$. Show that $\theta = \frac{64}{225}$ for $x = a$.

10. Show that the Cauchy's remainder after n terms in the expansion of $\ln(1+x)$ in powers of x is

$$(-1)^{n-1}\frac{x^n}{1+\theta x}\left(\frac{1-\theta}{1+\theta x}\right)^{n-1} \text{ where } 0<\theta<1.$$

11. Write the Maclaurin's series for the following functions:
(i) $y=\ln(1+\sin x)$ (ii) $y=\ln(1-\sin x)$ (iii) $y=\ln(1+\cos x)$

Chapter IV: Successive Differentiation, Leibniz Theorem

1. If $y=\dfrac{2x+3}{x^2+3x+2}$, then find y_n.

2. If $y=\dfrac{x^2}{(x-1)^2(x+2)}$, then find y_n.

3. If $y=\tan^{-1}\left(\dfrac{x}{a}\right)$, find y_n.

4. Find the n-th derivative of the following:
(i) $y=\cos x\cos 2x\cos 3x$ (ii) $y=e^{ax}\sin(bx+c)$ (iii) $y=e^{ax}\cos bx$
(iv) $y=e^{ax}\cos^2 x\sin x$ (v) $y=\cos^4 x$ (vi) $y=\cos^5 x$
(vii) $y=\sin^6 x$.

5. Given $x=\sin\left(\dfrac{\ln y}{m}\right)$. By applying Leibnitz's theorem prove that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0.$$

6. If $\cos^{-1}\left(\dfrac{y}{b}\right)=\ln\left(\dfrac{x}{n}\right)^n$ then prove that

$$x^2y_{n+2}+(2n+1)xy_{n+1}+2n^2y_n=0.$$

7. If $x=\sin\left(\dfrac{1}{m}\sin^{-1}y\right)$, then by applying Leibnitz's theorem prove that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0.$$

8. If $y=A\cos\{m\sin^{-1}(ax+b)\}$, then show that

$$\{1-(ax+b)^2\}y_{n+2}-(2n+1)a(ax+b)y_{n+1}+(m^2-n^2)a^2y_n=0.$$

9. If $y=(x^2-1)^n$, then show that

$$(x^2-1)y_{n+2}+2xy_{n+1}-n(n+1)y_n=0.$$

10. If $y=[x+\sqrt{1+x^2}]^m$, then show that

$$(1+x^2)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0.$$

11. If $y^{\frac{1}{m}}+y^{-\frac{1}{m}}=2x$, then show that

$$(x^2-1)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0.$$

12. If $y=\sin(m\sin^{-1}x)$, then by applying Leibnitz's theorem show that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0.$$

13. If $y=e^{m\cos^{-1}x}$, then by applying Leibnitz's theorem show that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(m^2+n^2)y_n=0.$$

14. If $y=a\cos(\ln x)+b\sin(\ln x)$, then by applying Leibnitz's theorem show that

(i) $x^2 y_2 + xy_1 + y = 0.$

(ii) $x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0.$

Chapter V: Maxima and Minima

1. Consider the following functions:

i. $y = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 6x + 8$ ii. $y = x^4 + 2x^3 - 3x^2 - 4x + 4$

iii. $y = 3x^4 - 10x^3 - 12x^2 + 12x - 7$ iv. $y = 3x^4 - 10x^3 - 12x^2 + 12x - 7$

v. $y = x^4 - 6x^3 + 12x^2 - 8x.$

Find the following:

(i) the critical points
(ii) the intervals on which y is increasing and decreasing
(iii) the maximum and minimum values of y
(iv) the points of inflection
(v) the equations of the tangents at the points of inflection

Finally, sketch the graph of $f(x)$.

2. Find the critical numbers of f if $f(x) = (x+5)^2 \sqrt[3]{x-4}.$

3. Given $f(x) = x^3 + x^2 - 5x - 5.$
   (i) Find the intervals on which f is increasing and the intervals on which f is decreasing.
   (ii) Find the intervals on which the graph of f is concave upward and concave downward.
   (iii) Sketch the graph of f.

4. If $f(x) = 12 + 2x^2 - x^4$, find the maximum and minimum values of f. Discuss concavity, find the points of inflection and sketch the graph of f.

5. Find the equations of the tangents at the points of inflection of
$$f(x) = x^4 - 6x^3 + 12x^2 - 8x$$
   Sketch the graph of the function.

6. Find the maximum volume of a right circular cylinder that can be inscribed in a cone of altitude 12 cm and base radius 4 cm, if the axes of the cylinder and cone coincide.

7. A cylindrical container with circular base is to hold 64 cubic inches. Find the dimensions so that the amount of metal required is a minimum.

8. A circular cylindrical container, open at the top and having a capacity of 24π cubic inches, is to be manufactured. If the cost of the material used for the bottom of the container is three times that used for the curved part and if there is no waste of material, find the dimensions which will minimize the cost.

9. Find the slopes of the lines that are parallel to *xz*- and *yz*-plane and tangent to the surface $z = 3x^2 + 4y^2 - 6$ at $P(1,3,2)$.

Chapter VI: Partial Differentiation

1. State Euler's theorem for homogeneous functions. If $u = \tan^{-1}\dfrac{5x^4 - 3xy^3 + 7y^2z^2}{x^2 - 2y^2 + 3yz}$ then show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = \sin 2u.$
2. If $u = \tan^{-1}\dfrac{x^2 + y^2 + z^2}{x + y + z}$ then show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = \dfrac{1}{2}\sin 2u.$
3. State Euler's theorem for homogeneous functions and verify the theorem for the function $f(x, y) = x^2 \log_e\left(\dfrac{y}{x}\right).$
4. If $u(x, y, z) = (x^2 + y^2 + z^2)^{\frac{-1}{2}}$, then show that u satisfies the Laplace's equation.
5. If $u(x, y, z) = 3e^{-2x}\cos 2y + x^3 - 3xy^2 + 5z - 7,$ then show that u satisfies the Laplace's equation.
6. If $u = \sin^{-1}\left(\dfrac{x}{y}\right) + \tan^{-1}\left(\dfrac{y}{x}\right),$ then find the value of $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y}.$
7. If $v = (x^2 + y^2 + z^2)^{\frac{m}{2}}$, then find the value of $\dfrac{\partial^2 v}{\partial x^2} + \dfrac{\partial^2 v}{\partial y^2} + \dfrac{\partial^2 v}{\partial z^2}.$
8. If $v = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta,$ then express $\dfrac{\partial v}{\partial x}$ and $\dfrac{\partial v}{\partial y}$ in polar coordinates.
9. If $u = f(x, y),$ where $x = r\cos\theta,\ y = r\sin\theta,$ prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

10. If $w = f\left(\dfrac{y - x}{xy}, \dfrac{z - y}{yz}\right),$ then prove that $x^2\dfrac{\partial w}{\partial x} + y^2\dfrac{\partial w}{\partial y} + z^2\dfrac{\partial w}{\partial z} = 0.$